JPX ENSEMBLE
Out-of-sample evaluation

Completed frozen experiment | 8 expanding folds | 160 training epochs

1. Executive assessment

Pooled MAE is **0.016060**, versus **0.015974** for a zero-return forecast. The relative MAE improvement is **-0.537%** (positive means lower error). Ranking and cost diagnostics are separate from prediction error.

Metric	Pooled out-of-sample value
Evaluated stock-days / trading dates	1,853,400 / 958
Validation Huber loss (delta = 1)	0.00029055
Sharpe, unannualized	0.0768
Sharpe x sqrt(252), IID reference only	1.2189
Mean daily Rank IC / defined dates	0.0190 / 548 of 958
Gross additive spread-return sum	17.781%
Constant-prediction dates	409 of 958
Illustrative break-even cost per traded notional	3.132 bps

These outputs do **not** establish deployable alpha. Constant forecasts can make selected holdings depend on tie ordering; the mean Rank IC omits undefined dates. Costs and execution constraints must be considered before interpreting gross spreads as an investable strategy.

Recorded active compute: 16.16 hours. Elapsed wall time: 20.10 hours, including pauses. Start: 2026-10-17T07:55:52.817932+00:00; finish: 2026-10-18T04:02:05.642923+00:00 (UTC).

Includes successful preprocessing, training, validation and checkpoint overhead; abandoned partial epochs after abrupt interruption may not be counted.

2. Fold results and coverage

Each completed fold uses 20 training epochs followed by chronological out-of-sample validation. Model and Adam state carry into subsequent folds; these are not independent model fits.

Fold	MAE	Zero MAE	MAE gain	Huber loss	Sharpe*	Rank IC
1	0.012171	0.012123	-0.40%	0.0001845	0.0417	0.0105
2	0.014975	0.014707	-1.83%	0.0002564	0.0158	0.0092
3	0.017460	0.017017	-2.61%	0.0003233	-0.0451	0.0162
4	0.015815	0.015819	0.03%	0.0002794	0.0194	0.0774
5	0.013341	0.013376	0.26%	0.0001977	0.1228	0.0375
6	0.022400	0.022472	0.32%	0.0005305	0.1059	0.0610
7	0.016934	0.016914	-0.12%	0.0003077	-0.0259	0.0164
8	0.015336	0.015328	-0.06%	0.0002453	0.3328	0.0190

* Sharpe is unannualized. MAE gain = (zero MAE - model MAE) / zero MAE. Fold portfolio metrics preserve the original float32 reductions; pooled reductions use float64.

Fold	Validation dates	Stock-days	Dates	Constant forecasts	Undefined IC
1	2017-06-30 to 2017-12-26	228,969	122	2	2
2	2017-12-27 to 2018-06-22	225,375	119	6	6
3	2018-06-25 to 2018-12-21	238,566	125	31	31
4	2018-12-25 to 2019-06-19	219,599	114	100	100
5	2019-06-20 to 2019-12-16	234,905	121	82	82
6	2019-12-17 to 2020-06-12	229,702	117	93	93
7	2020-06-15 to 2020-12-10	239,011	121	63	64
8	2020-12-11 to 2021-06-08	237,273	119	32	32

Undefined daily IC is excluded from the average, not replaced by zero. A high reported IC over a small subset of dates must not be generalized to every trading day. Constant forecasts can still produce nonzero portfolio returns because the original code assigns holdings after sorting ties.

3. Baselines and comparison scope

The zero-return baseline predicts Target = 0 for the same ensemble validation stock-days. Pooled MAE and Huber loss are weighted by validation sample count, not averaged equally across folds.

The historical CNN-LSTM comparison uses **958 common dates**. Stock counts differ on 683 of those dates. Same-date alignment is therefore not identical-security alignment.

Common-date metric	ENSEMBLE	CNN-LSTM
Unannualized Sharpe	0.0768	0.0994
Gross additive spread sum	17.781%	23.595%
Mean daily spread (bps)	1.856	2.463
Mean Rank IC	0.0190	0.0038
Defined IC dates	548	957

Design choice	ENSEMBLE	CNN-LSTM
Lookback	21 trading observations	42 trading observations
Training loss	Huber, delta 1	L1 / MAE
Architecture	GRU + RNN + LSTM hidden-state mean	CNN-LSTM
Random initialization	Seed 20261017	Different seed

This is a comparison of two experimental configurations, **not a controlled architecture-only experiment**. Different lookbacks, losses, seeds and sample coverage can explain differences. No claim of statistical significance or live profitability is made.

4. Trading costs and return definitions

The original portfolio selects 200 long and 200 short stocks. Rank weights taper from 2 to 1, normalized separately to **+0.5 long and -0.5 short**. Daily spread = sum(weight x Target). Original earnings = sum(daily spreads): an **uncompounded, pre-cost additive sum**, not CAGR and not a compounded net equity return.

Primary Sharpe = mean(daily spread) / sample standard deviation (ddof = 1), with zero base return. The separately labeled sqrt(252) annualization is an IID reference only; it does not correct for autocorrelation or establish an executable holding-period interpretation.

Cost (bps)*	Additive scenario sum	Mean / date (bps)	Sharpe, unannualized
0	17.781%	1.856	0.0768
1	12.103%	1.263	0.0524
2	6.425%	0.671	0.0279
5	-10.608%	-1.107	-0.0459
10	-38.996%	-4.071	-0.1641

* 1 basis point (bp) = 0.01%. Costs are charged per absolute traded target notional; this is not a fixed daily cost or an assumed real broker quotation.

Turnover convention: sum of absolute target-weight changes across securities, starting flat, carrying holdings across folds and adding the final close. Cost on a date = bps x 0.0001 x turnover. Total modeled traded target notional = 567.774. No half-turnover convention is used.

Execution limitations: JPX Target uses the return from the next close to the following close. Prediction date, data availability, entry time, overlapping holdings and rebalance accounting need explicit reconciliation in an execution simulator. These scenarios ignore price drift, slippage, bid-ask spreads, borrow availability/fees, taxes, liquidity and market impact. They are sensitivity calculations, **not realized net performance**.

5. Frozen protocol, limitations and audit

Component	Recorded protocol
Input and source	Uploaded codes.zip; nine original Python files kept byte-for-byte intact. Uploaded stock_prices.csv.
Network	GRU, RNN and LSTM; hidden 128, one layer each; average final hidden states; dropout 0.2; dense 128 -> 32 -> 1.
Training	Six features; lookback 21; batch 64; no shuffle; float32; Huber delta 1; Adam default lr 0.001; no scheduler.
State and compute	CPU, four Torch threads and one interop thread; seed 20261017. Model and Adam continue across folds.
Recovery	Atomic epoch/fold checkpoints retain model, Adam, scheduler=null, Python/NumPy/Torch RNG, scaler/context, losses, progress and prior validation.

Interpretation limits. Historical sample selection and available lookback history restrict the universe. Scaling is fitted before the two-date purge: the purge does not remove all preprocessing dependence on those dates. Expanding folds share training history and optimizer state. Ties use average ranks followed by the original sort behavior, so tied selections may depend on ordering. A single seed and no untouched final test set limit robustness claims.

Audit scope. Per-fold predictions and metrics are reconstructed from saved artifacts. A final report requires eight folds, 160 epochs, valid final-artifact hashes, and audited final weights/scaler agreement with the last checkpoint. Small eager/lazy and interrupted-resume equivalence tests validate implementation continuity; they do not validate investment performance.

CSV SHA256:
bf774a86f834e5338bba74f2356ebb750f93a1cc4e89b064ee07251e8ed851db

Analysis summary SHA256:
8ed1b1e54421e24f431d7bea8a0417c98913dcf8f9985492747ce6226e91f84b

Supporting details: cloud_run/input_hashes.json; requirements.txt; results/fold_results.json; per-fold prediction files; BACKUP_MANIFEST.json; reports/final_analysis tables; final model and resumable checkpoint in the result bundle.

JPX competition reference: Evaluation | Data / Target definition. This is an offline source-code evaluation, not a verified Kaggle submission score.